Multivariate Data Analysis

Part II: Reduction, Classification and Clustering (Ch. 7–11)

Contents

Chapter 7. Principal Component Analysis 4

1 Population Principal Components 4

2 Sample Principal Components and the SVD 5

2.1 Sample PCs . . . 5

2.2 Relation to the singular value decomposition . . . 5

3 PCA and the Best Approximating Affine Subspace 6

3.1 The problem . . . 6

3.2 Observations . . . 6

3.3 Correlation PCA . . . 8

4 How Many Components to Keep? 8

5 Example: Scores in Two Subjects Revisited 9

Chapter 8. Canonical Correlation Analysis 10

1 Setting 10

2 Solution via the Singular Value Decomposition 10

3 Sample Canonical Correlations 11

3.1 Regularized CCA . . . 11

4 Testing Canonical Correlations 11

5 Relations to PCA, Multiple Correlation and Discriminant Analysis 12

Chapter 9. Factor Analysis 13

1 The Orthogonal Factor Model 13

1.1 Non-identifiability and rotation . . . 13

1.2 Invariance to scale changes . . . 13

2 Estimation **14**
2.1 Principal component method . . . 14
2.2 Principal factor method . . . 14
2.3 Maximum likelihood . . . 14
2.4 Likelihood ratio test for the number of factors . . . 17

3 Factor Rotation **17**

4 Factor Score Prediction **17**
4.1 Weighted least squares (Bartlett) scores . . . 17
4.2 Regression scores . . . 17

5 PCA versus Factor Analysis **18**

Chapter 10. Discrimination and Classification **19**

1 The Bayes Classification Rule **19**
1.1 Setting . . . 19

2 Normal Populations: LDA and QDA **19**
2.1 Common covariance: linear discriminant analysis . . . 19
2.2 Different covariances: quadratic discriminant analysis . . . 20

3 Fisher's Linear Discriminant **20**

4 Logistic Regression **21**

5 Simple Classifiers in High Dimensions **21**

6 Assessment of Classifiers **22**

Chapter 11. Cluster Analysis **23**

1 Dissimilarity and Within-Cluster Scatter **23**
1.1 Limits of combinatorial algorithms . . . 23

2 K-means and K-medoids **24**
2.1 K-means . . . 24
2.2 K-medoids . . . 24

3 Hierarchical Clustering **24**
3.1 Agglomerative . . . 24
3.2 Divisive . . . 25

4 Gaussian Mixtures and the EM Algorithm **25**

4.1 Model 25

4.2 The EM algorithm ($K = 2$) 25

5 Choosing K and Other Topics **26**

Chapter 7. Principal Component Analysis

Population and sample principal components, SVD, choosing the number of components

Ch. 1, §6, introduced PCA through sample geometry. Here it is formulated at the population level, with the Lagrange-multiplier derivation, the SVD and the choice of the number of components.

1 Population Principal Components

Motivation. With many correlated variables, most of the variation often lies in a few directions. Principal components are the orthogonal directions of maximal variance, the eigenvectors of $\boldsymbol{\Sigma}$, and they allow us to summarize the data in fewer dimensions.

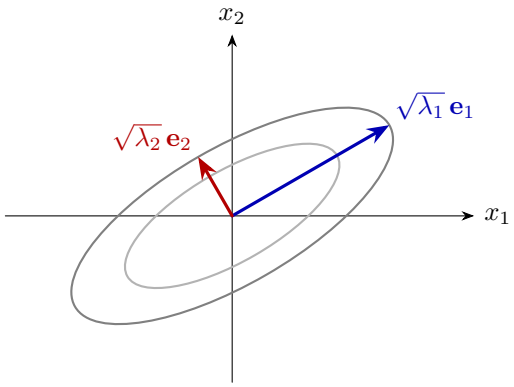


Figure. *Contours of a bivariate normal density are ellipses whose axes are the eigenvectors $\mathbf{e}_1, \mathbf{e}_2$ of $\boldsymbol{\Sigma}$, with half-lengths proportional to $\sqrt{\lambda_1}, \sqrt{\lambda_2}$. The first principal component is the projection on the long axis, the direction of maximal variance.*

Let $\mathbf{X} \in \mathbb{R}^p$ with $\mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$, $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$. A linear combination $Z(\mathbf{a}) = \mathbf{a}^\top \mathbf{X}$ has variance $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$. (For $d = 1$: $\mathbf{a}_1 = \arg\max_{\|\mathbf{a}\|=1} \mathrm{Var}(Z(\mathbf{a}))$.)

Definition 7.1: principal components

First PC direction vector: $\mathbf{u}_1 = \arg\max_{\|\mathbf{u}\|=1} \mathrm{Var}(Z(\mathbf{u}))$, with $\lambda_1 = \mathrm{Var}(Z_1)$. Second: $\mathbf{u}_2 = \arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_1 = 0} \mathrm{Var}(\mathbf{u}^\top \mathbf{X})$, with $\lambda_2 = \mathrm{Var}(Z_2)$; and so on:

$$\mathbf{u}_k = \arg \max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_j = 0 \ (j<k)} \mathrm{Var}(\mathbf{u}^\top \mathbf{X}), \qquad Z_k = \mathbf{u}_k^\top \mathbf{X}.$$

Then $\lambda_1 \geq \lambda_2 \geq \cdots$ and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Theorem 7.2: solution

$\mathbf{u}_k$ is the kth eigenvector of $\boldsymbol{\Sigma}$, $\mathrm{Var}(Z_k) = \lambda_k$, and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Lagrange multipliers [Supplement]. **1.** $\Phi(\mathbf{u}, \lambda) = \mathrm{Var}(\mathbf{u}^\top \mathbf{X}) - \lambda(\mathbf{u}^\top \mathbf{u} - 1) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1)$. $\frac{1}{2} \frac{\partial}{\partial \mathbf{u}} \Phi(\mathbf{u}, \lambda) = \boldsymbol{\Sigma}\mathbf{u} - \lambda \mathbf{u} = \mathbf{0}$, so $\boldsymbol{\Sigma}\mathbf{u} = \lambda \mathbf{u}$ and $\lambda = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u}$: the objective equals the eigenvalue, so take the largest, λ_1.

2. With $\boldsymbol{\gamma}^k = (\gamma_1, \dots, \gamma_{k-1})$,

$$\Phi(\mathbf{u}, \lambda, \boldsymbol{\gamma}^k) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1) - \sum_{j=1}^{k-1} 2\gamma_j \mathbf{u}_j^\top \mathbf{u}, \qquad \frac{1}{2} \frac{\partial \Phi}{\partial \mathbf{u}} = \boldsymbol{\Sigma}\mathbf{u} - \lambda \mathbf{u} - \sum_{j=1}^{k-1} \gamma_j \mathbf{u}_j = \mathbf{0}, \quad \frac{\partial \Phi}{\partial \gamma_j} = \mathbf{u}_j^\top \mathbf{u} = 0.$$

Premultiplying by $\mathbf{u}_i^\top$ $(i<k)$: $\mathbf{u}_i^\top\boldsymbol{\Sigma}\mathbf{u}=\gamma_i$, and $\mathbf{u}_i^\top\boldsymbol{\Sigma}\mathbf{u}=\lambda_i\mathbf{u}_i^\top\mathbf{u}=0$, so $\gamma_i=0$. Hence $\boldsymbol{\Sigma}\mathbf{u}=\lambda\mathbf{u}$ again, and among eigenvectors orthogonal to $\mathbf{u}_1,\dots,\mathbf{u}_{k-1}$ the largest eigenvalue λ_k is chosen: $\mathbf{u}_k,\lambda_k$. Finally $\mathrm{Cov}(Z_j,Z_k)=\mathbf{u}_j^\top\boldsymbol{\Sigma}\mathbf{u}_k=\lambda_k\mathbf{u}_j^\top\mathbf{u}_k=0$. ■

Theorem 7.3: total variance and correlations

1. $\sum_{j=1}^p \mathrm{Var}(X_j)=\mathrm{tr}\boldsymbol{\Sigma}=\sum_{k=1}^p\lambda_k=\sum_k\mathrm{Var}(Z_k)$; the proportion of variance explained by PC k is $\lambda_k/\sum_j\lambda_j$.
2. $\mathrm{Corr}(Z_k,X_j)=\dfrac{u_{jk}\sqrt{\lambda_k}}{\sqrt{\sigma_{jj}}}$, where u_{jk} is the jth entry of $\mathbf{u}_k$.

Proof [Supplement]. (1) $\mathrm{tr}(\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top)=\mathrm{tr}\boldsymbol{\Lambda}$.
(2) $\mathrm{Cov}(Z_k,X_j)=\mathbf{u}_k^\top\boldsymbol{\Sigma}\mathbf{e}_j=\lambda_k\mathbf{u}_k^\top\mathbf{e}_j=\lambda_k u_{jk}$, so $\mathrm{Corr}=\lambda_k u_{jk}/(\sqrt{\lambda_k}\sqrt{\sigma_{jj}})$. ■

Theorem 7.4: Ky Fan maximum principle — q-dimensional subspaces

$$\max_{\mathbf{V}\in\mathbb{R}^{p\times q},\ \mathbf{V}^\top\mathbf{V}=\mathbf{I}_q}\mathrm{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V})=\lambda_1+\cdots+\lambda_q,$$

attained at $\mathbf{V}=[\mathbf{u}_1,\dots,\mathbf{u}_q]$ (and its rotations).

Proof [Supplement]. $\mathbf{P}=\mathbf{V}\mathbf{V}^\top$ is a rank-q projection with eigenvalues 1 (q times) and 0. By von Neumann's upper bound, $\mathrm{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V})=\mathrm{tr}(\boldsymbol{\Sigma}\mathbf{P})\le\sum_i\lambda_i(\boldsymbol{\Sigma})\lambda_i(\mathbf{P})=\lambda_1+\cdots+\lambda_q$. ■

Remark 7.5: PCA as optimal approximation

$\mathrm{E}\|(\mathbf{X}-\boldsymbol{\mu})-\mathbf{V}\mathbf{V}^\top(\mathbf{X}-\boldsymbol{\mu})\|^2=\mathrm{tr}\boldsymbol{\Sigma}-\mathrm{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V})$, so among q-dimensional subspaces through the mean, the first q PCs minimize the expected squared reconstruction error, the minimum being $\lambda_{q+1}+\cdots+\lambda_p$.

2 Sample Principal Components and the SVD

2.1 Sample PCs

Let $\tilde{\mathbf{X}}=\mathbf{H}_n\mathbf{X}=\mathbf{X}-\mathbf{1}_n\bar{\mathbf{x}}^\top$ be the centered data matrix, $\mathbf{S}=\frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}=\hat{\mathbf{U}}\hat{\boldsymbol{\Lambda}}\hat{\mathbf{U}}^\top$, and

$$\hat{\mathbf{u}}_k=\arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top\hat{\mathbf{u}}_j=0,\ j=1,\dots,k-1}\mathbf{u}^\top\mathbf{S}\mathbf{u},\qquad \text{scores } t_{ik}=\hat{\mathbf{u}}_k^\top(\mathbf{x}_i-\bar{\mathbf{x}}).$$

The sample PC scores have sample variance $\hat{\lambda}_k$ and are mutually uncorrelated.

2.2 Relation to the singular value decomposition

Write the SVD of the $p\times n$ matrix $\tilde{\mathbf{X}}^\top$ $(n>p)$ as

$$\tilde{\mathbf{X}}^\top=\mathbf{U}\mathbf{D}\mathbf{V}^\top=\underbrace{[\mathbf{u}_1\cdots\mathbf{u}_p]}_{p\times p}\underbrace{\begin{pmatrix}d_1&&\\&\ddots&\\&&d_p\end{pmatrix}}_{p\times p}\underbrace{\begin{pmatrix}\mathbf{v}_1^\top\\\vdots\\\mathbf{v}_p^\top\end{pmatrix}}_{p\times n},\qquad \mathbf{U}^\top\mathbf{U}=\mathbf{I}_p,\ \mathbf{V}^\top\mathbf{V}=\mathbf{I}_p,\ d_1\ge\cdots\ge d_p\ge0.$$

Then

$$\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \mathbf{U}\mathrm{diag}\Big(\frac{d_1^2}{n-1},\dots,\frac{d_p^2}{n-1}\Big)\mathbf{U}^\top, \qquad \begin{pmatrix}\mathbf{z}_{(1)}^\top\\ \vdots \\ \mathbf{z}_{(p)}^\top\end{pmatrix} = \mathbf{Z} = \mathbf{U}^\top\tilde{\mathbf{X}}^\top = \mathbf{D}\mathbf{V}^\top = \begin{pmatrix} d_1\mathbf{v}_1^\top \\ \vdots \\ d_p\mathbf{v}_p^\top\end{pmatrix}.$$

The PC directions are the left singular vectors, the eigenvalues are $d_k^2/(n-1)$, and the kth PC score vector is $\mathbf{z}_{(k)} = \hat{\mathbf{u}}_k^\top\tilde{\mathbf{X}}^\top = d_k\mathbf{v}_k^\top$. Numerically the SVD is preferred to forming $\mathbf{S}$ (computation of PCA).

3 PCA and the Best Approximating Affine Subspace

Motivation. A second view of PCA: the first r components span the r-dimensional affine subspace closest to the data in squared distance. Maximizing variance and minimizing reconstruction error are the same problem.

Theorem 7.6: claim

PCA with r components is equivalent to finding the best r-dimensional affine subspace (hyperplane) approximating the data.

3.1 The problem

Let the data be $\mathbf{x}_1,\dots,\mathbf{x}_n \in \mathbb{R}^p$ and $r \in \{1,\dots,p\}$. We wish to approximate $\mathbf{x}_1,\dots,\mathbf{x}_n$ by $\bar{\mathbf{x}}+\mathbf{a}_1,\dots,\bar{\mathbf{x}}+\mathbf{a}_n$ where $\mathrm{rank}([\mathbf{a}_1,\dots,\mathbf{a}_n]) = r$. The approximation of $\mathbf{x}_i$ by $\bar{\mathbf{x}}+\mathbf{a}_i$ is better when the error $\|\mathbf{x}_i - (\bar{\mathbf{x}}+\mathbf{a}_i)\|_2^2$ is smaller, and the approximation of $\{\mathbf{x}_1,\dots,\mathbf{x}_n\}$ by $\{\bar{\mathbf{x}}+\mathbf{a}_i\}$ is better when $\mathrm{SSE} = \sum_{i=1}^n \|\mathbf{x}_i - (\bar{\mathbf{x}}+\mathbf{a}_i)\|_2^2$ is smaller. That is, we wish to find $\mathbf{A}_r = [\mathbf{a}_1,\dots,\mathbf{a}_n]$ $(p\times n)$ solving

$$(\star) \qquad \min_{\mathbf{A}_r}\sum_{i=1}^n \|(\mathbf{x}_i-\bar{\mathbf{x}})-\mathbf{a}_i\|_2^2 \quad \text{subject to} \quad \mathrm{rank}(\mathbf{A}_r) = r.$$

Since every column of a rank-r matrix lies in an r-dimensional subspace, the points $\bar{\mathbf{x}}+\mathbf{a}_i$ lie on an r-dimensional affine subspace through $\bar{\mathbf{x}}$; conversely any such subspace gives feasible $\mathbf{A}_r$.

3.2 Observations

1. With $\tilde{\mathbf{X}}^\top = [\mathbf{x}_1-\bar{\mathbf{x}},\dots,\mathbf{x}_n-\bar{\mathbf{x}}]$ $(p\times n)$,

$$\sum_{i=1}^n \|(\mathbf{x}_i-\bar{\mathbf{x}})-\mathbf{a}_i\|_2^2 = \|\tilde{\mathbf{X}}^\top - \mathbf{A}_r\|_F^2,$$

because for a $p\times n$ matrix $\mathbf{B} = [\mathbf{b}_1,\dots,\mathbf{b}_n]$ the Frobenius norm is $\|\mathbf{B}\|_F^2 = \sum_{i=1}^p\sum_{j=1}^n b_{ij}^2 = \sum_{j=1}^n\|\mathbf{b}_j\|_2^2 = \mathrm{tr}(\mathbf{B}\mathbf{B}^\top) = \mathrm{tr}(\mathbf{B}^\top\mathbf{B})$.

2. The sample PCA of $(\mathbf{x}_1,\dots,\mathbf{x}_n)$ is given by

   (1) the eigendecomposition $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top = \sum_{i=1}^p \hat{\lambda}_i\hat{\mathbf{u}}_i\hat{\mathbf{u}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{\lambda}_i$ the ith PC variance, $\hat{\lambda}_1 \ge \cdots \ge \hat{\lambda}_r > \hat{\lambda}_{r+1} \ge \cdots \ge \hat{\lambda}_p$;

   (2) the SVD $\tilde{\mathbf{X}}_{(p\times n)}^\top = \hat{\mathbf{U}}\hat{\mathbf{D}}\hat{\mathbf{V}}^\top = \sum_{i=1}^p \hat{d}_i\hat{\mathbf{u}}_i\hat{\mathbf{v}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{d}_i\hat{\mathbf{v}}_i$ the ith PC score vector;

(3) $\hat{\mathbf{u}}_i^\top \tilde{\mathbf{X}}^\top = \hat{d}_i \hat{\mathbf{v}}_i^\top$, i.e.

$$\begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix}.$$

Theorem 7.7: solution (Eckart–Young–Mirsky)

The $\mathbf{A}_r$ that minimizes $(\star)$ is

$$\mathbf{A}_r = \sum_{i=1}^{r} \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r][\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r]^\top \tilde{\mathbf{X}}^\top,$$

the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace, and the minimum is $\sum_{i>r} \hat{d}_i^2 = (n-1)\sum_{i=r+1}^{p} \hat{\lambda}_i$.

Proof [Supplement]. **Step 1 (parametrize rank-r matrices).** Since $\operatorname{rank}(\mathbf{A}_r) = r$, we can write $\mathbf{A}_r = \mathbf{E}\mathbf{C}^\top$ with $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_r]$ $(p \times r)$ satisfying $\mathbf{E}^\top \mathbf{E} = \mathbf{I}_r$ (an orthonormal basis of the column space of $\mathbf{A}_r$) and $\mathbf{C} = \begin{pmatrix} \mathbf{c}_1^\top \\ \vdots \\ \mathbf{c}_n^\top \end{pmatrix}$ an $n \times r$ matrix. Writing $\tilde{\mathbf{x}}_i = \mathbf{x}_i - \bar{\mathbf{x}}$, $(\star)$ is equivalent to $\min_{\mathbf{E},\mathbf{C}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2$, where

$$(\star\star) \qquad \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2 = \big\|(\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_n) - \mathbf{E}[\mathbf{c}_1, \dots, \mathbf{c}_n]\big\|_F^2 = \sum_{i=1}^{n} \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2.$$

Step 2 (optimal $\mathbf{C}$ for fixed $\mathbf{E}$). For any given $\mathbf{E}$, $(\star\star)$ separates over i and its minimizer is

$$\hat{\mathbf{c}}_i = \arg\min_{\mathbf{c}_i} \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2 = (\mathbf{E}^\top \mathbf{E})^{-1} \mathbf{E}^\top \tilde{\mathbf{x}}_i = \mathbf{E}^\top \tilde{\mathbf{x}}_i,$$

the projection scores of $\tilde{\mathbf{x}}_i$ onto $\operatorname{span}(\mathbf{E})$ (least squares). Stacking, $\hat{\mathbf{C}} = \begin{pmatrix} \tilde{\mathbf{x}}_1^\top \mathbf{E} \\ \vdots \\ \tilde{\mathbf{x}}_n^\top \mathbf{E} \end{pmatrix} = \tilde{\mathbf{X}}\mathbf{E}$, i.e. $\hat{\mathbf{C}}^\top = \mathbf{E}^\top \tilde{\mathbf{X}}^\top$.

Step 3 (profile out $\mathbf{C}$). Plugging $\hat{\mathbf{C}}$ into $(\star\star)$ (profiling, Theorem 2.22), and using that $\mathbf{I} - \mathbf{E}\mathbf{E}^\top$ is symmetric idempotent, $(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)^2 = \mathbf{I} - \mathbf{E}\mathbf{E}^\top$,

$$\begin{aligned} \min_{\mathbf{E}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{E}^\top \tilde{\mathbf{X}}^\top\|_F^2 &= \min_{\mathbf{E}} \|(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top\|_F^2 = \min_{\mathbf{E}} \operatorname{tr}(\tilde{\mathbf{X}}(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top) \\ &= \min_{\mathbf{E}} \Big\{ \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1)\operatorname{tr}\Big(\mathbf{E}^\top \frac{\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}}{n-1} \mathbf{E}\Big) \Big\} \\ &= \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1) \max_{\mathbf{E}} \operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top). \end{aligned}$$

Step 4 (von Neumann). Recall $\mathbf{S} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top$. $\mathbf{E}\mathbf{E}^\top$ is symmetric with eigenvalues 1 (r times) and 0 ($p - r$ times), so von Neumann's trace inequality (Theorem 2.12) gives

$$\operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top) \le \hat{\lambda}_1 + \cdots + \hat{\lambda}_r,$$

with equality when $\mathbf{E} = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r]$ (or any $\mathbf{E}$ spanning the same space).

Step 5 (the minimizer). Therefore

$$\mathbf{A}_r = \mathbf{E}\hat{\mathbf{C}}^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix} = \sum_{i=1}^{r} \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top$$

by Observation 2(3): the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace. Since $\mathrm{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) = (n-1)\mathrm{tr}\mathbf{S} = (n-1)\sum_i \hat{\lambda}_i$, the minimum is $(n-1)\sum_{i>r}\hat{\lambda}_i = \sum_{i>r}\hat{d}_i^2$. ■

Remark 7.8: interpretation

- The best approximating hyperplane is $\bar{\mathbf{x}}+\mathrm{span}\{\hat{\mathbf{u}}_1,\dots,\hat{\mathbf{u}}_r\}$ and each $\mathbf{x}_i$ is approximated by its orthogonal projection $\bar{\mathbf{x}} + \sum_{k\le r}(\hat{\mathbf{u}}_k^\top \tilde{\mathbf{x}}_i)\hat{\mathbf{u}}_k$: the PC scores are the coordinates on the hyperplane.
- The fitted values satisfy $\hat{\mathbf{c}}_i = \mathbf{E}^\top \tilde{\mathbf{x}}_i$, i.e. reconstruction uses projection scores, exactly the decomposition $\mathbf{x} = \bar{\mathbf{x}} + \sum s_{\mathbf{v}_j}(\mathbf{x}-\bar{\mathbf{x}})\mathbf{v}_j + r_{\mathcal{L}_q}(\mathbf{x})$ of Ch. 1.
- Minimizing squared error and maximizing retained variance (the $\arg\max \mathbf{u}^\top\mathbf{S}\mathbf{u}$ definition) are the same problem, since $\mathrm{SSE} = (n-1)\{\mathrm{tr}\mathbf{S} - \mathrm{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top)\}$.
- If $\hat{\lambda}_r = \hat{\lambda}_{r+1}$ the optimal subspace is not unique; the strict gap $\hat{\lambda}_r > \hat{\lambda}_{r+1}$ makes it unique.

3.3 Correlation PCA

If the units differ, use PCA of standardized data: $\mathbf{R} = \mathbf{D}_{\mathbf{\Sigma}}^{-1/2}\mathbf{\Sigma}\mathbf{D}_{\mathbf{\Sigma}}^{-1/2} = \mathbf{U}_R\mathbf{\Lambda}_R\mathbf{U}_R^\top$ (sample: $\mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2}$). Since $\mathrm{tr}\mathbf{R} = p$, the proportion explained is λ_k^R/p. Covariance and correlation PCs are in general not simple transformations of each other (scale dependence).

4 How Many Components to Keep?

- **Scree plot / cumulative scree plot**: plot $\hat{\lambda}_k$ against k and look for the elbow, or take the q at which the cumulative proportion reaches a target (e.g. 80–90%).
- **Kaiser's rule**: keep components with $\hat{\lambda}_k > \bar{\lambda} = \frac{1}{p}\sum_j \hat{\lambda}_j$; for correlation PCA, $\hat{\lambda}_k > 1$.
- **Likelihood ratio testing**: under normality test $H_0(k) : \lambda_{k+1} = \cdots = \lambda_p$ (the remaining space is spherical). With arithmetic mean $\bar{a}$ and geometric mean $\bar{g}$ of the last $p-k$ sample eigenvalues,

$$-2\log\Lambda = n(p-k)\log\frac{\bar{a}}{\bar{g}} \;\dot{\sim}\; \chi^2_{(p-k+2)(p-k-1)/2};$$

since $\bar{a} \ge \bar{g}$ (AM–GM), the statistic is small when the eigenvalues are nearly equal.

Remark 7.9: cautions in interpretation

- The sign of an eigenvector is arbitrary; what matters is the sign pattern (same signs: "size"; mixed signs: "contrast"). In the scores example $\mathbf{u}_1 \propto (1,1)$ was overall level and $\mathbf{u}_2 \propto (1,-1)$ the contrast between subjects.
- With tied eigenvalues the directions within the eigenspace are not unique.
- Sample eigenvalues are biased: large ones are overestimated and small ones underestimated. Under normality, for distinct λ_k, $\sqrt{n}(\hat{\lambda}_k - \lambda_k) \Rightarrow N(0, 2\lambda_k^2)$.
- High explained variance does not mean usefulness for classification or prediction (the $X_1 = 10Z$, $X_2 = Y$ example of Ch. 1).

5 Example: Scores in Two Subjects Revisited

For $\mathbf{S} = \begin{pmatrix} 125 & 75 \\ 75 & 125 \end{pmatrix}$ the characteristic equation $(125-\lambda)^2 - 75^2 = 0$ gives $\lambda = 200, 50$; the null space of $\mathbf{S} - 200\mathbf{I} = \begin{pmatrix} -75 & 75 \\ 75 & -75 \end{pmatrix}$ is spanned by $(1,1)^\top$. $\mathrm{Corr}(Z_1, X_1) = u_{11}\sqrt{\lambda_1}/\sqrt{s_{11}} = \frac{1}{\sqrt{2}}\sqrt{200}/\sqrt{125} = 0.894$, $\mathrm{Corr}(Z_2, X_1) = \frac{1}{\sqrt{2}}\sqrt{50}/\sqrt{125} = 0.447$, and $0.894^2 + 0.447^2 = 1$: each variable's variance is fully decomposed over the PCs.

Chapter 8. Canonical Correlation Analysis

Maximal correlation between two sets of variables

1 Setting

We want to summarize the association between two sets of variables $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$ with $\operatorname{Cov}\begin{pmatrix}\mathbf{X}\\ \mathbf{Y}\end{pmatrix} = \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{pmatrix}$, $\boldsymbol{\Sigma}_{11}, \boldsymbol{\Sigma}_{22} \succ 0$. Instead of inspecting pq correlations, maximize the correlation of linear combinations

$$\xi = \mathbf{g}^\top \mathbf{X}, \qquad \omega = \mathbf{h}^\top \mathbf{Y}.$$

For $q = 1$ this is the multiple correlation (Ch. 3). PCA maximizes variance *within* one set; CCA maximizes correlation *between* two sets.

Definition 8.1: canonical variates and canonical correlations

$$(\mathbf{g}_1, \mathbf{h}_1) = \arg \max_{\mathbf{g}\in\mathbb{R}^p,\, \mathbf{h}\in\mathbb{R}^q} \operatorname{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}),$$

$$(\mathbf{g}_k, \mathbf{h}_k) = \arg \max_{\substack{\mathbf{g}\in\mathbb{R}^p,\, \mathbf{h}\in\mathbb{R}^q \\ \mathbf{g}^\top \boldsymbol{\Sigma}_{11} \mathbf{g}_j = 0,\ \mathbf{h}^\top \boldsymbol{\Sigma}_{22} \mathbf{h}_j = 0,\ j=1,\dots,k-1}} \operatorname{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}).$$

$\xi_k = \mathbf{g}_k^\top \mathbf{X}$, $\omega_k = \mathbf{h}_k^\top \mathbf{Y}$ form the kth pair of canonical variates and $\rho_k = \operatorname{Corr}(\xi_k, \omega_k)$ is the kth canonical correlation. Correlation is scale-free, so normalize $\operatorname{Var}(\xi_k) = \operatorname{Var}(\omega_k) = 1$. The constraints mean $\operatorname{Cov}(\xi_k, \xi_j) = \operatorname{Cov}(\omega_k, \omega_j) = 0$, and $\rho_k \le \rho_j$ for $j < k$.

$\rho(\mathbf{X}, \mathbf{Y}) := \max \operatorname{Corr}(\mathbf{a}^\top \mathbf{X}, \mathbf{b}^\top \mathbf{Y}) = \rho_1$ is the **largest canonical coefficient**.

2 Solution via the Singular Value Decomposition

Motivation. The low-rank approximation problem is solved directly by the SVD (Eckart–Young), which also gives a numerically stable way to compute principal components.

Theorem 8.2: solution of CCA

Let $\mathbf{M} = \boldsymbol{\Sigma}_{11}^{-1/2} \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1/2}$ $(p \times q)$ have SVD $\mathbf{M} = \sum_{k=1}^{r} d_k \mathbf{a}_k \mathbf{b}_k^\top$, $d_1 \ge \cdots \ge d_r > 0$, $r = \operatorname{rank} \boldsymbol{\Sigma}_{12}$. Then

$$\rho_k = d_k, \qquad \mathbf{g}_k = \boldsymbol{\Sigma}_{11}^{-1/2} \mathbf{a}_k, \qquad \mathbf{h}_k = \boldsymbol{\Sigma}_{22}^{-1/2} \mathbf{b}_k.$$

Equivalently $\mathbf{a}_k$ is the kth eigenvector of $\mathbf{M}\mathbf{M}^\top = \boldsymbol{\Sigma}_{11}^{-1/2} \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1/2}$, $\mathbf{b}_k$ that of $\mathbf{M}^\top \mathbf{M} = \boldsymbol{\Sigma}_{22}^{-1/2} \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1/2}$, and ρ_k^2 is the eigenvalue.

Proof [Supplement]. Substituting $\mathbf{a} = \boldsymbol{\Sigma}_{11}^{1/2} \mathbf{g}$, $\mathbf{b} = \boldsymbol{\Sigma}_{22}^{1/2} \mathbf{h}$,

$$\operatorname{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}) = \frac{\mathbf{g}^\top \boldsymbol{\Sigma}_{12} \mathbf{h}}{\sqrt{\mathbf{g}^\top \boldsymbol{\Sigma}_{11} \mathbf{g}} \sqrt{\mathbf{h}^\top \boldsymbol{\Sigma}_{22} \mathbf{h}}} = \frac{\mathbf{a}^\top \mathbf{M} \mathbf{b}}{\sqrt{\mathbf{a}^\top \mathbf{a}} \sqrt{\mathbf{b}^\top \mathbf{b}}},$$

so we maximize $\mathbf{a}^\top \mathbf{M} \mathbf{b}$ over $\|\mathbf{a}\| = \|\mathbf{b}\| = 1$. With $\mathbf{M} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$, $\boldsymbol{\alpha} = \mathbf{U}^\top \mathbf{a}$, $\boldsymbol{\beta} = \mathbf{V}^\top \mathbf{b}$, Cauchy–Schwarz gives

$$\mathbf{a}^\top \mathbf{M} \mathbf{b} = \mathbf{a}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top \mathbf{b} = \sum_k d_k \alpha_k \beta_k \le d_1 \sum_k |\alpha_k \beta_k| \le d_1 \|\boldsymbol{\alpha}\| \|\boldsymbol{\beta}\| \le d_1,$$

with equality at $\mathbf{a} = \mathbf{a}_1$, $\mathbf{b} = \mathbf{b}_1$. Under $\mathbf{g}^\top \boldsymbol{\Sigma}_{11} \mathbf{g}_j = \mathbf{a}^\top \mathbf{a}_j = 0$ and $\mathbf{b}^\top \mathbf{b}_j = 0$ $(j < k)$ we have $\alpha_j = \beta_j = 0$ for $j < k$, and the same argument gives maximum d_k at $(\mathbf{a}_k, \mathbf{b}_k)$. ■

Theorem 8.3: properties of canonical variates

1. $\mathrm{Var}(\xi_k) = \mathrm{Var}(\omega_k) = 1$ and $\mathrm{Corr}(\xi_k, \xi_l) = \mathrm{Corr}(\omega_k, \omega_l) = 0$ for $k \neq l$.
2. $\mathrm{Corr}(\xi_k, \omega_l) = 0$ for $k \neq l$, $\mathrm{Corr}(\xi_k, \omega_k) = \rho_k$.
3. Invariance: canonical correlations are unchanged by $\mathbf{X} \mapsto \mathbf{D}_1 \mathbf{X} + \mathbf{c}_1$, $\mathbf{Y} \mapsto \mathbf{D}_2 \mathbf{Y} + \mathbf{c}_2$ with $\mathbf{D}_i$ non-singular; covariance and correlation matrices give the same ρ_k.
4. For $q = 1$, $\rho_1^2 = \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\Sigma}_{12} / \sigma_{22}$ is the squared multiple correlation.

Proof [Supplement]. (1) $\mathrm{Cov}(\xi_k, \xi_l) = \mathbf{g}_k^\top \boldsymbol{\Sigma}_{11} \mathbf{g}_l = \mathbf{a}_k^\top \mathbf{a}_l = \delta_{kl}$.

(2) $\mathrm{Cov}(\xi_k, \omega_l) = \mathbf{g}_k^\top \boldsymbol{\Sigma}_{12} \mathbf{h}_l = \mathbf{a}_k^\top \mathbf{M} \mathbf{b}_l = d_l \mathbf{a}_k^\top \mathbf{a}_l = d_l \delta_{kl}$.

(3) $\mathrm{Corr}(\mathbf{g}^\top \mathbf{D}_1 \mathbf{X}, \mathbf{h}^\top \mathbf{D}_2 \mathbf{Y}) = \mathrm{Corr}((\mathbf{D}_1^\top \mathbf{g})^\top \mathbf{X}, (\mathbf{D}_2^\top \mathbf{h})^\top \mathbf{Y})$ and $\mathbf{g} \mapsto \mathbf{D}_1^\top \mathbf{g}$ is a bijection, so the maxima agree.

(4) For $q = 1$ the only nonzero eigenvalue of $\mathbf{M}\mathbf{M}^\top$ is $\mathbf{M}^\top \mathbf{M} = \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\Sigma}_{12} / \sigma_{22}$. ■

3 Sample Canonical Correlations

Replace the covariances by $\mathbf{S}_{11} = \widehat{\mathrm{Cov}}(\mathbf{X})$, $\mathbf{S}_{22} = \widehat{\mathrm{Cov}}(\mathbf{Y})$, $\mathbf{S}_{12} = \widehat{\mathrm{Cov}}(\mathbf{X}, \mathbf{Y})$:

$$(\hat{\mathbf{g}}_k, \hat{\mathbf{h}}_k) = \arg \max_{\substack{\mathbf{g} \in \mathbb{R}^p,\, \mathbf{h} \in \mathbb{R}^q \\ \mathbf{g}^\top \mathbf{S}_{11} \hat{\mathbf{g}}_j = 0,\ \mathbf{h}^\top \mathbf{S}_{22} \hat{\mathbf{h}}_j = 0,\ j = 1, \ldots, k-1}} \frac{\mathbf{g}^\top \mathbf{S}_{12} \mathbf{h}}{\sqrt{\mathbf{g}^\top \mathbf{S}_{11} \mathbf{g}\, \mathbf{h}^\top \mathbf{S}_{22} \mathbf{h}}}.$$

With $\mathbf{a} = \mathbf{S}_{11}^{1/2} \mathbf{g}$, $\mathbf{b} = \mathbf{S}_{22}^{1/2} \mathbf{h}$ this is maximizing $\dfrac{\mathbf{a}^\top \mathbf{S}_{11}^{-1/2} \mathbf{S}_{12} \mathbf{S}_{22}^{-1/2} \mathbf{b}}{\sqrt{\mathbf{a}^\top \mathbf{a}\, \mathbf{b}^\top \mathbf{b}}} = \max_{\|\mathbf{a}\| = \|\mathbf{b}\| = 1} \mathbf{a}^\top \mathbf{M} \mathbf{b} = \mathbf{a}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top \mathbf{b}$, so

$$\mathbf{a}_i = \mathbf{v}_i(\mathbf{S}_{11}^{-1/2} \mathbf{S}_{12} \mathbf{S}_{22}^{-1} \mathbf{S}_{21} \mathbf{S}_{11}^{-1/2}), \qquad \mathbf{b}_i = \mathbf{v}_i(\mathbf{S}_{22}^{-1/2} \mathbf{S}_{21} \mathbf{S}_{11}^{-1} \mathbf{S}_{12} \mathbf{S}_{22}^{-1/2}),$$

where $\mathbf{v}_i(\mathbf{M})$ is the ith eigenvector of $\mathbf{M}$.

3.1 Regularized CCA

If p or q is large relative to n, $\mathbf{S}_{11}, \mathbf{S}_{22}$ are singular or unstable and sample canonical correlations are biased towards 1. **Regularized CCA** adds a ridge term,

$$\mathbf{S}_{11} \mapsto \mathbf{S}_{11} + \lambda_1 \mathbf{I}, \qquad \mathbf{S}_{22} \mapsto \mathbf{S}_{22} + \lambda_2 \mathbf{I},$$

with λ_1, λ_2 chosen by cross-validation.

4 Testing Canonical Correlations

Under multivariate normality the likelihood ratio statistic for $H_0 : \boldsymbol{\Sigma}_{12} = \mathbf{0}$ (all $\rho_k = 0$) is

$$\Lambda^{2/n} = \frac{|\mathbf{S}|}{|\mathbf{S}_{11}||\mathbf{S}_{22}|} = \prod_{k=1}^{\min(p,q)} (1 - \hat{\rho}_k^2),$$

and Bartlett's approximation gives $-\big(n-1-\frac{p+q+1}{2}\big)\sum_k \ln(1-\hat\rho_k^2) \dot\sim \chi^2_{pq}$. The hypothesis that only the first k are nonzero, $H_0^{(k)}: \rho_{k+1}=\cdots=0$, is tested sequentially with

$$-\Big(n-1-\frac{p+q+1}{2}\Big)\sum_{i=k+1}^{\min(p,q)} \ln(1-\hat\rho_i^2) \;\dot\sim\; \chi^2_{(p-k)(q-k)}.$$

Derivation: the determinant identity

By the block determinant formula, $|\mathbf{S}| = |\mathbf{S}_{22}||\mathbf{S}_{11}-\mathbf{S}_{12}\mathbf{S}_{22}^{-1}\mathbf{S}_{21}| = |\mathbf{S}_{22}||\mathbf{S}_{11}|\,|\mathbf{I}-\hat{\mathbf{M}}\hat{\mathbf{M}}^\top|$, and the eigenvalues of $\hat{\mathbf{M}}\hat{\mathbf{M}}^\top$ are $\hat\rho_k^2$, so $|\mathbf{S}|/(|\mathbf{S}_{11}||\mathbf{S}_{22}|) = \prod(1-\hat\rho_k^2)$.

5 Relations to PCA, Multiple Correlation and Discriminant Analysis

- **Multiple correlation**: CCA with $q=1$.
- **Discriminant analysis**: with $\mathbf{Y}$ a set of group indicator (dummy) variables, the canonical variates of $\mathbf{X}$ give the same directions as Fisher's discriminants (Ch. 10).
- **PCA**: variance maximization within one set (eigendecomposition of $\mathbf{\Sigma}$) versus correlation maximization between two sets (SVD of $\mathbf{\Sigma}_{11}^{-1/2}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1/2}$).

Chapter 9. Factor Analysis

Model, estimation (principal components, principal factors, MLE), rotation, factor scores

1 The Orthogonal Factor Model

Motivation. Factor analysis explains correlations by a few unobserved common factors: $\mathbf{\Sigma} = \mathbf{L}\mathbf{L}^\top + \mathbf{\Psi}$. Unlike PCA, it is a model with specific error variances, and the loadings are determined only up to rotation.

The covariance of p observed variables is explained by a few, k, unobserved common factors.

Definition 9.1: factor model

$$X_i = \mu_i + \sum_{\ell=1}^{k} \lambda_{i\ell} F_\ell + U_i \quad (i = 1, \dots, p), \qquad \mathbf{X} = \boldsymbol{\mu} + \mathbf{\Lambda}\mathbf{F} + \mathbf{U},$$

where $\mathbf{\Lambda} = (\lambda_{i\ell})$ is the $p \times k$ **loading matrix**, $\mathbf{F} \in \mathbb{R}^k$ the common factors and $\mathbf{U} \in \mathbb{R}^p$ the specific factors, with

$$\mathrm{E}(\mathbf{F}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{F}) = \mathbf{I}_k, \quad \mathrm{E}(\mathbf{U}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{U}) = \mathbf{\Psi} = \mathrm{diag}(\psi_1, \dots, \psi_p), \quad \mathrm{Cov}(\mathbf{F}, \mathbf{U}) = \mathbf{0}.$$

Theorem 9.2: covariance structure

$$\mathrm{E}(\mathbf{X}) = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\mathbf{X}) = \mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi}, \qquad \mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \mathbf{\Lambda},$$

$$\mathrm{Var}(X_i) = \underbrace{\sum_{\ell=1}^{k} \lambda_{i\ell}^2}_{h_i^2\text{: communality}} + \underbrace{\psi_i}_{\text{specific variance}}, \qquad \mathrm{Cov}(X_i, X_j) = \sum_{\ell} \lambda_{i\ell}\lambda_{j\ell} \ (i \neq j).$$

Proof [Supplement]. $\mathrm{Cov}(\mathbf{\Lambda}\mathbf{F} + \mathbf{U}) = \mathbf{\Lambda}\mathrm{Cov}(\mathbf{F})\mathbf{\Lambda}^\top + \mathrm{Cov}(\mathbf{U}) + \mathbf{\Lambda}\mathrm{Cov}(\mathbf{F}, \mathbf{U}) + \mathrm{Cov}(\mathbf{U}, \mathbf{F})\mathbf{\Lambda}^\top = \mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi}$, and $\mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \mathrm{Cov}(\mathbf{\Lambda}\mathbf{F} + \mathbf{U}, \mathbf{F}) = \mathbf{\Lambda}$. ■

Because $\mathbf{\Psi}$ is diagonal, covariances between variables come *entirely* from the common factors. The model explains the $p(p+1)/2$ entries of $\mathbf{\Sigma}$ with $pk + p$ parameters.

1.1 Non-identifiability and rotation

For orthogonal $\mathbf{T}$ $(k \times k)$, $\mathbf{\Lambda}^* = \mathbf{\Lambda}\mathbf{T}$ and $\mathbf{F}^* = \mathbf{T}^\top\mathbf{F}$ give $\mathbf{\Lambda}^*\mathbf{F}^* = \mathbf{\Lambda}\mathbf{F}$, $\mathrm{Cov}\mathbf{F}^* = \mathbf{I}_k$ and $\mathbf{\Lambda}^*\mathbf{\Lambda}^{*\top} = \mathbf{\Lambda}\mathbf{\Lambda}^\top$: loadings are determined only up to rotation. If $\hat{\mathbf{\Lambda}}$ is an MLE, so is $\hat{\mathbf{\Lambda}}\mathbf{R}$ for any orthogonal $\mathbf{R}$. Removing the $k(k-1)/2$ rotational degrees of freedom leaves $pk + p - k(k-1)/2$ parameters and

$$\nu = \frac{p(p+1)}{2} - \left\{pk + p - \frac{k(k-1)}{2}\right\} = \frac{(p-k)^2 - p - k}{2},$$

which must be positive for the model to impose a real restriction.

1.2 Invariance to scale changes

If $Y_i = c_i X_i$, i.e. $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\mathbf{X}$, then $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\boldsymbol{\mu} + \mathrm{diag}(\mathbf{c})\mathbf{\Lambda}\mathbf{F} + \mathrm{diag}(\mathbf{c})\mathbf{U}$ is again a factor model with $\mathbf{\Lambda}_Y = \mathrm{diag}(\mathbf{c})\mathbf{\Lambda}$ and $\mathbf{\Psi}_Y = \mathrm{diag}(\mathbf{c})^2\mathbf{\Psi}$. For standardized variables ($\mathrm{Var}X_i = 1$),

$$\lambda_{i\ell} = \mathrm{Cov}(X_i, F_\ell) = \mathrm{Corr}(X_i, F_\ell), \qquad \mathrm{Corr}(Y_i, F_\ell) = \lambda_{i\ell},$$

so loadings are read as variable–factor correlations. In practice factor analysis is often run on the correlation matrix $\mathbf{R}$.

2 Estimation

2.1 Principal component method

From $\widehat{\mathrm{Corr}}(\mathbf{X}) = \mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} = \sum_{j=1}^{p} l_j \mathbf{u}_j \mathbf{u}_j^\top \approx \sum_{j=1}^{k} l_j \mathbf{u}_j \mathbf{u}_j^\top$ (largest k eigenvalues),

$$\hat{\mathbf{\Lambda}} = [\sqrt{l_1}\mathbf{u}_1, \ldots, \sqrt{l_k}\mathbf{u}_k], \qquad \hat{\mathbf{\Psi}} = \mathrm{diag}(\mathbf{R} - \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top).$$

The diagonal of $\mathbf{R}$ is reproduced exactly, and the sum of squared off-diagonal residuals of $\mathbf{R} - \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top - \hat{\mathbf{\Psi}}$ is at most $l_{k+1}^2 + \cdots + l_p^2$. Factor ℓ explains a proportion l_ℓ/p of the total variance. Since eigenvalues do not change, increasing k leaves earlier loadings unchanged.

Remark 9.3: number of factors with the principal component method

Heuristics are used when PCA is used to estimate the factor loadings:

1. use a **scree plot** or a **cumulative scree plot** of $l_1 \geq l_2 \geq \cdots \geq l_p$;
2. or use **Kaiser's rule**: retain factors $1, \ldots, k$ satisfying

$$l_k > \bar{l} = \frac{1}{p}\sum_{i=1}^{p} l_i.$$

When correlation PCA is used, $\sum_i l_i = \mathrm{tr}\mathbf{R} = p$, so $\bar{l} = 1$ and the rule keeps factors with $l_k > 1$ (a factor must explain more than one standardized variable's worth of variance).

These are the same rules as for choosing the number of principal components (Ch. 7); the likelihood ratio test of §2.4 is available only under the normal MLE approach.

2.2 Principal factor method

Starting from initial communality estimates $\hat{h}_i^2$ (e.g. $1 - 1/r^{ii}$, r^{ii} the diagonal of $\mathbf{R}^{-1}$), form the reduced correlation matrix $\mathbf{R} - \hat{\mathbf{\Psi}}$ and iterate

1. $\mathbf{R} - \hat{\mathbf{\Psi}} \approx \hat{\mathbf{\Lambda}}_{(k)}\hat{\mathbf{\Lambda}}_{(k)}^\top$ (top-k eigendecomposition of the reduced matrix);
2. $\hat{\psi}_{(k)i} = 1 - \sum_{\ell=1}^{k} \hat{\lambda}_{(k)i\ell}^2$

until convergence.

2.3 Maximum likelihood

Assume $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})$. Profiling $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, twice the negative log-likelihood is, up to constants and a factor n,

$$\ell(\hat{\boldsymbol{\mu}}, \mathbf{\Lambda}, \mathbf{\Psi}) = \log|\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi}| + \mathrm{tr}\{\mathbf{S}_0(\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})^{-1}\} + c.$$

For identifiability impose that $\mathbf{\Delta} = \mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda}$ is diagonal (if $\hat{\mathbf{\Lambda}}$ is an MLE, so is $\hat{\mathbf{\Lambda}}\mathbf{R}$).

Theorem 9.4: (∗) $\boldsymbol{\Lambda}$ given $\boldsymbol{\Psi}$

Given $\boldsymbol{\Psi} = \mathrm{diag}(\psi_1, \dots, \psi_p)$, let $\mathbf{A} = \boldsymbol{\Psi}^{-1/2}\mathbf{S}_0\boldsymbol{\Psi}^{-1/2} = \sum_{i=1}^{p} d_i \mathbf{e}_i \mathbf{e}_i^\top$ $(d_1 \geq \cdots \geq d_p)$ and $\delta_i = d_i - 1 > 0$ $(i \leq k)$. Then

$$\hat{\boldsymbol{\Lambda}} = \boldsymbol{\Psi}^{1/2}[\mathbf{e}_1, \dots, \mathbf{e}_k]\mathrm{diag}(\delta_1^{1/2}, \dots, \delta_k^{1/2}).$$

Proof [Supplement]. Let $\boldsymbol{\Lambda}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}$ and $\boldsymbol{\Sigma}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Sigma}\boldsymbol{\Psi}^{-1/2} = \boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} + \mathbf{I}_p$. Then $\ell = \log|\boldsymbol{\Psi}| + \log|\boldsymbol{\Sigma}^*| + \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1})$. If $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top}$ has eigenvalues $\gamma_1 \geq \cdots \geq \gamma_k \geq 0 = \cdots = 0$, $\boldsymbol{\Sigma}^*$ has eigenvalues $1+\gamma_j$ and 1. Von Neumann's lower bound (pair the small eigenvalues of $\boldsymbol{\Sigma}^{*-1}$ with the large ones of $\mathbf{A}$) gives

$$\log|\boldsymbol{\Sigma}^*| + \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1}) \geq \sum_{j=1}^{k}\Big\{\log(1+\gamma_j) + \frac{d_j}{1+\gamma_j}\Big\} + \sum_{j>k} d_j,$$

with equality when the eigenvectors of $\boldsymbol{\Sigma}^*$ coincide with the $\mathbf{e}_j$ in the same order. Each term $\log x + d/x = \log d + \{d/x - \log(d/x)\} \geq \log d + 1$ (Theorem 2.13) is minimized at $x = d$, so $1+\gamma_j = d_j$, i.e. $\gamma_j = \delta_j$. Hence $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} = \sum_{j\leq k}\delta_j \mathbf{e}_j\mathbf{e}_j^\top$ and $\boldsymbol{\Lambda}^* = [\mathbf{e}_1, \dots, \mathbf{e}_k]\mathrm{diag}(\delta_j^{1/2})$ up to rotation; this choice makes $\boldsymbol{\Lambda}^{*\top}\boldsymbol{\Lambda}^* = \boldsymbol{\Delta}$ diagonal. ■

Theorem 9.5: (∗∗) $\boldsymbol{\Psi}$ given $\boldsymbol{\Lambda}$

The MLE satisfies $\mathrm{diag}(\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}) = \mathrm{diag}(\mathbf{S}_0)$ (the model reproduces the sample variances). Accordingly, given $\boldsymbol{\Lambda}$, update

$$\hat{\boldsymbol{\Psi}} = \mathrm{diag}(\hat{\psi}_1, \dots, \hat{\psi}_p), \qquad \hat{\psi}_i = (\mathbf{S}_0 - \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top)_{ii}.$$

Alternate the two steps. If some $\hat{\psi}_i \leq 0$ (a Heywood case), bound it below by a small positive number.

Logic for (∗) and (∗∗): the likelihood equations

We evaluate $\dfrac{\partial \ell(\boldsymbol{\Lambda}, \boldsymbol{\Psi})}{\partial \boldsymbol{\Psi}} = 0$ and $\dfrac{\partial \ell(\boldsymbol{\Lambda}, \boldsymbol{\Psi})}{\partial \boldsymbol{\Lambda}} = 0$ for

$$\ell(\boldsymbol{\Lambda}, \boldsymbol{\Psi}) = \log|\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}| + \mathrm{tr}\{\mathbf{S}_0(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1}\}, \qquad \boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi},$$

using the facts that for a matrix $\mathbf{A} = \mathbf{A}(x)$,

$$\frac{\partial \mathbf{A}^{-1}}{\partial x} = -\mathbf{A}^{-1}\frac{\partial \mathbf{A}}{\partial x}\mathbf{A}^{-1}, \qquad \frac{\partial \log|\mathbf{A}|}{\partial x} = \mathrm{tr}\Big(\mathbf{A}^{-1}\frac{\partial \mathbf{A}}{\partial x}\Big)$$

(the second is the scalar form of Theorem 2.19; the first follows from differentiating $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$).

Derivative in ψ_i. With $\mathbf{e}_i$ the ith unit vector, $\frac{\partial}{\partial \psi_i}\mathrm{diag}(\psi_1, \dots, \psi_p) = \mathbf{e}_i\mathbf{e}_i^\top$, so $\frac{\partial \boldsymbol{\Sigma}}{\partial \psi_i} = \mathbf{e}_i\mathbf{e}_i^\top$ and

$$\frac{\partial \ell}{\partial \psi_i} = \underbrace{\mathrm{tr}\Big(\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \psi_i}\Big)}_{\partial \log|\boldsymbol{\Sigma}|/\partial\psi_i} + \underbrace{\mathrm{tr}\Big(\mathbf{S}_0\frac{\partial \boldsymbol{\Sigma}^{-1}}{\partial \psi_i}\Big)}_{\partial \mathrm{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1})/\partial\psi_i} = \mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{e}_i^\top) - \mathrm{tr}(\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{e}_i\mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1})$$
$$= \mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{e}_i - \mathbf{e}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}\mathbf{e}_i = \mathbf{e}_i^\top(\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1})\mathbf{e}_i = 0, \qquad i = 1, \dots, p.$$

$$\textcircled{1} \qquad \mathrm{diag}(\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}) = \mathbf{0}.$$

Derivative in λ_{ij}. With $\mathbf{\Lambda} = [\mathbf{l}_1, \dots, \mathbf{l}_k]$ (columns), $\dfrac{\partial \mathbf{\Lambda}\mathbf{\Lambda}^\top}{\partial \lambda_{ij}} = \mathbf{l}_j \mathbf{e}_i^\top + \mathbf{e}_i \mathbf{l}_j^\top$. Then

$$\begin{aligned}\frac{\partial \ell}{\partial \lambda_{ij}} &= \operatorname{tr}\Big(\mathbf{\Sigma}^{-1}\frac{\partial \mathbf{\Lambda}\mathbf{\Lambda}^\top}{\partial \lambda_{ij}}\Big) - \operatorname{tr}\Big(\mathbf{S}_0\mathbf{\Sigma}^{-1}\frac{\partial \mathbf{\Sigma}}{\partial \lambda_{ij}}\mathbf{\Sigma}^{-1}\Big) \\ &= \operatorname{tr}(\mathbf{\Sigma}^{-1}\mathbf{l}_j\mathbf{e}_i^\top) + \operatorname{tr}(\mathbf{\Sigma}^{-1}\mathbf{e}_i\mathbf{l}_j^\top) - \operatorname{tr}(\mathbf{S}_0\mathbf{\Sigma}^{-1}\mathbf{l}_j\mathbf{e}_i^\top\mathbf{\Sigma}^{-1}) - \operatorname{tr}(\mathbf{S}_0\mathbf{\Sigma}^{-1}\mathbf{e}_i\mathbf{l}_j^\top\mathbf{\Sigma}^{-1}) \\ &= 2\mathbf{e}_i^\top\mathbf{\Sigma}^{-1}\mathbf{l}_j - 2\mathbf{e}_i^\top\mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1}\mathbf{l}_j = 0 \qquad (i = 1, \dots, p,\ j = 1, \dots, k),\end{aligned}$$

using symmetry of $\mathbf{\Sigma}^{-1}$ and $\mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1}$ (each pair of traces is equal). Collecting over i, j:

$$\mathbf{\Sigma}^{-1}\mathbf{\Lambda} = \mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1}\mathbf{\Lambda} \overset{\text{premultiply } \mathbf{\Sigma}}{\Longrightarrow} \mathbf{\Lambda} = \mathbf{S}_0\mathbf{\Sigma}^{-1}\mathbf{\Lambda} = \mathbf{S}_0(\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})^{-1}\mathbf{\Lambda} = \mathbf{S}_0\mathbf{\Psi}^{-1}\mathbf{\Lambda}(\mathbf{I} + \mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda})^{-1},$$

where the last step uses $\mathbf{\Lambda}(\mathbf{I} + \mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda}) = (\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})\mathbf{\Psi}^{-1}\mathbf{\Lambda}$, i.e. $(\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})^{-1}\mathbf{\Lambda} = \mathbf{\Psi}^{-1}\mathbf{\Lambda}(\mathbf{I} + \mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda})^{-1}$.

Write $\mathbf{\Delta} \equiv \mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda}$ (and require $\mathbf{\Delta}_{k\times k}$ to be diagonal). Then

$$②' \qquad \mathbf{\Lambda}(\mathbf{I} + \mathbf{\Delta}) = \mathbf{S}_0\mathbf{\Psi}^{-1}\mathbf{\Lambda}.$$

Premultiplying by $\mathbf{\Psi}^{-1/2}$:

$$② \qquad \underbrace{\mathbf{\Psi}^{-1/2}\mathbf{S}_0\mathbf{\Psi}^{-1/2}}_{=:\mathbf{A}}\ \underbrace{\mathbf{\Psi}^{-1/2}\mathbf{\Lambda}}_{=:\tilde{\mathbf{E}}} = \underbrace{\mathbf{\Psi}^{-1/2}\mathbf{\Lambda}}_{\tilde{\mathbf{E}}}\ \underbrace{(\mathbf{I} + \mathbf{\Delta})}_{=:\mathbf{D}=\operatorname{diag}(d_1,\dots,d_k)}, \qquad \tilde{\mathbf{E}} = [\tilde{\mathbf{e}}_1, \dots, \tilde{\mathbf{e}}_k].$$

Hence $\mathbf{A}\tilde{\mathbf{e}}_i = \tilde{\mathbf{e}}_i d_i$, $i = 1, \dots, k$: $(\tilde{\mathbf{e}}_i, d_i)$ is an eigenpair of $\mathbf{A}$, although $\tilde{\mathbf{e}}_i^\top\tilde{\mathbf{e}}_i \neq 1$ in general. Since $\tilde{\mathbf{e}}_i^\top\tilde{\mathbf{e}}_i = (\tilde{\mathbf{E}}^\top\tilde{\mathbf{E}})_{ii} = (\mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda})_{ii} = \Delta_{ii}$, let $\mathbf{e}_i \equiv \tilde{\mathbf{e}}_i/\Delta_{ii}^{1/2}$ so that $\mathbf{e}_i^\top\mathbf{e}_i = 1$ and, for $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_k]$, $\mathbf{E}^\top\mathbf{E} = \mathbf{I}_k$ ($\mathbf{\Delta}$ diagonal). Then $(\mathbf{e}_i, d_i)$ is a unit eigenvector–eigenvalue pair of $\mathbf{A}$, and

$$\mathbf{D} = \mathbf{I} + \mathbf{\Delta}, \qquad \tilde{\mathbf{E}} = \mathbf{\Psi}^{-1/2}\mathbf{\Lambda} = \mathbf{E}\mathbf{\Delta}^{1/2}.$$

($*$) $\mathbf{\Psi}$ is fixed. From ②, $\mathbf{A} = \mathbf{\Psi}^{-1/2}\mathbf{S}_0\mathbf{\Psi}^{-1/2}$ is computable, with eigenpairs $(\mathbf{e}_i(\mathbf{A}), d_i(\mathbf{A}))$. Use $\mathbf{D} = \mathbf{I} + \mathbf{\Delta}$ to set $\hat{\mathbf{\Delta}} = \operatorname{diag}(d_i(\mathbf{A})) - \mathbf{I}$, and $\mathbf{E}\mathbf{\Delta}^{1/2} = \mathbf{\Psi}^{-1/2}\mathbf{\Lambda}$ to set

$$\hat{\mathbf{\Lambda}} = \mathbf{\Psi}^{1/2}\mathbf{E}(\mathbf{A})\hat{\mathbf{\Delta}}^{1/2}, \qquad \mathbf{E}(\mathbf{A}) = [\mathbf{e}_1(\mathbf{A}), \dots, \mathbf{e}_k(\mathbf{A})],$$

taking the k largest eigenvalues (the von Neumann argument above shows these give the minimum).

($**$) $\mathbf{\Lambda}$ is fixed. From ①, $\operatorname{diag}(\mathbf{\Sigma}^{-1} - \mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1}) = \mathbf{0}$.

Claim: if ②′ holds, then

$$\mathbf{\Sigma}^{-1} - \mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1} = \mathbf{\Sigma}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Sigma}^{-1} = \mathbf{\Psi}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1} = \mathbf{\Psi}^{-1}(\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{S}_0 + \mathbf{\Psi})\mathbf{\Psi}^{-1}.$$

Proof of claim [Supplement]. $\mathbf{\Sigma}\mathbf{\Psi}^{-1}\mathbf{\Lambda} = (\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})\mathbf{\Psi}^{-1}\mathbf{\Lambda} = \mathbf{\Lambda}(\mathbf{I} + \mathbf{\Delta})$, and by ②′ also $\mathbf{S}_0\mathbf{\Psi}^{-1}\mathbf{\Lambda} = \mathbf{\Lambda}(\mathbf{I} + \mathbf{\Delta})$, so $(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1}\mathbf{\Lambda} = \mathbf{0}$. By Woodbury, $\mathbf{\Sigma}^{-1} = \mathbf{\Psi}^{-1} - \mathbf{\Psi}^{-1}\mathbf{\Lambda}(\mathbf{I} + \mathbf{\Delta})^{-1}\mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}$, hence

$$(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Sigma}^{-1} = (\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1} - \underbrace{(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1}\mathbf{\Lambda}}_{=\mathbf{0}}(\mathbf{I} + \mathbf{\Delta})^{-1}\mathbf{\Lambda}^\top\mathbf{\Psi}^{-1} = (\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1},$$

and transposing, $\mathbf{\Sigma}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0) = \mathbf{\Psi}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)$. Therefore $\mathbf{\Sigma}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Sigma}^{-1} = \mathbf{\Sigma}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1} = \mathbf{\Psi}^{-1}(\mathbf{\Sigma} - \mathbf{S}_0)\mathbf{\Psi}^{-1}$.

Since $\mathbf{\Psi}^{-1}$ is diagonal, the ith diagonal entry of $\mathbf{\Psi}^{-1}(\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{S}_0 + \mathbf{\Psi})\mathbf{\Psi}^{-1}$ is $(\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{S}_0 + \mathbf{\Psi})_{ii}/\psi_i^2$, so ① becomes

$$\operatorname{diag}(\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{S}_0 + \mathbf{\Psi}) = \mathbf{0} \quad \Longrightarrow \quad \psi_i = (\mathbf{S}_0 - \mathbf{\Lambda}\mathbf{\Lambda}^\top)_{ii},$$

which is the update ($**$) and also shows $\operatorname{diag}(\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}}) = \operatorname{diag}(\mathbf{S}_0)$ at the MLE.

2.4 Likelihood ratio test for the number of factors

$H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$ (k factors) versus H_1: unrestricted $\boldsymbol{\Sigma}$. With $L_1 = L(\bar{\mathbf{x}}, \mathbf{S}_0)$ and $L_0 = L(\bar{\mathbf{x}}, \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}})$,

$$W = -2\log\frac{L_0}{L_1} = n\log\Big(\frac{|\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}|}{|\mathbf{S}_0|}\Big) \;\dot{\sim}\; \chi^2_\nu,$$

$$\nu = \frac{p(p+1)}{2} - \Big\{pk + p - \frac{k(k-1)}{2}\Big\} = \frac{(p-k)^2 - p - k}{2}.$$

(At the MLE $\operatorname{tr}\{\mathbf{S}_0\hat{\boldsymbol{\Sigma}}^{-1}\} = p$, so the trace terms cancel.) With Bartlett's correction, reject the k-factor model if $(n - 1 - \frac{2p+4k+5}{6}) \ln\frac{|\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}|}{|\mathbf{S}_0|} > \chi^2_\nu(\alpha)$.

3 Factor Rotation

Non-identifiability is used to find interpretable loadings, ideally a simple structure in which each variable loads heavily on only one factor. Rotate $\boldsymbol{\Lambda}^* = \boldsymbol{\Lambda}\mathbf{J}$ with $\mathbf{J}$ orthogonal.

Definition 9.6: varimax and quartimax

Let $\tilde{\lambda}^*_{i\ell} = \lambda^*_{i\ell}/h_i$ (normalized by the communality).

$$\text{Varimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{\ell=1}^{k} \widehat{\operatorname{Var}}\big((\tilde{\lambda}^*_{1\ell})^2, \ldots, (\tilde{\lambda}^*_{p\ell})^2\big)$$

— spread the squared loadings within each *column* (factor), so each factor has some large and some near-zero loadings.

$$\text{Quartimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{i=1}^{p} \widehat{\operatorname{Var}}\big((\lambda^*_{i1})^2, \ldots, (\lambda^*_{ik})^2\big)$$

— spread the squared loadings within each *row* (variable), so each variable concentrates on one factor.

Rotation leaves the communalities h_i^2 and $\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top$ unchanged.

4 Factor Score Prediction

4.1 Weighted least squares (Bartlett) scores

(1) Choose $\hat{\mathbf{F}}$ to minimize the distance between $\mathbf{X}$ and $\boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F}$: with the estimates fixed, regard $\mathbf{x}_i - \bar{\mathbf{x}} = \hat{\boldsymbol{\Lambda}}\mathbf{F}_i + \mathbf{U}_i$ as a heteroscedastic regression,

$$\hat{\mathbf{F}}_i^{\text{wls}} = \arg\min_{\mathbf{F}_i} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\boldsymbol{\Lambda}}\mathbf{F}_i)^\top \hat{\boldsymbol{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\boldsymbol{\Lambda}}\mathbf{F}_i) = (\hat{\boldsymbol{\Lambda}}^\top \hat{\boldsymbol{\Psi}}^{-1} \hat{\boldsymbol{\Lambda}})^{-1} \hat{\boldsymbol{\Lambda}}^\top \hat{\boldsymbol{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$$

(set the gradient of the quadratic form to zero).

4.2 Regression scores

(2) Since $\begin{pmatrix}\mathbf{X}\\ \mathbf{F}\end{pmatrix} \sim N_{p+k}\Big(\begin{pmatrix}\boldsymbol{\mu}\\ \mathbf{0}\end{pmatrix}, \begin{pmatrix}\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi} & \boldsymbol{\Lambda}\\ \boldsymbol{\Lambda}^\top & \mathbf{I}_k\end{pmatrix}\Big)$, the conditional expectation (Theorem 3.11) gives

$$\hat{\mathbf{F}}_i^{\text{reg}} = \mathrm{E}(\mathbf{F} \mid \mathbf{X} = \mathbf{x}_i) = \hat{\boldsymbol{\Lambda}}^\top (\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}})^{-1} (\mathbf{x}_i - \bar{\mathbf{x}}).$$

Theorem 9.7: relation between the two scores

With $\boldsymbol{\Delta} = \boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$, $\hat{\mathbf{F}}^{\text{reg}} = (\mathbf{I}+\boldsymbol{\Delta})^{-1}\boldsymbol{\Delta}\,\hat{\mathbf{F}}^{\text{wls}} = (\mathbf{I}+\boldsymbol{\Delta}^{-1})^{-1}\hat{\mathbf{F}}^{\text{wls}}$. If $\boldsymbol{\Delta}$ is diagonal each component is shrunk by $\delta_\ell/(1+\delta_\ell) < 1$, so $|\hat{F}_\ell^{\text{wls}}| \geq |\hat{F}_\ell^{\text{reg}}|$.

Proof [Supplement]. By Woodbury, $(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1} = \boldsymbol{\Psi}^{-1} - \boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}(\mathbf{I}+\boldsymbol{\Delta})^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}$, so

$$\boldsymbol{\Lambda}^\top(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1} = \{\mathbf{I} - \boldsymbol{\Delta}(\mathbf{I}+\boldsymbol{\Delta})^{-1}\}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1} = (\mathbf{I}+\boldsymbol{\Delta})^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}$$

(as $\mathbf{I} - \boldsymbol{\Delta}(\mathbf{I}+\boldsymbol{\Delta})^{-1} = (\mathbf{I}+\boldsymbol{\Delta}-\boldsymbol{\Delta})(\mathbf{I}+\boldsymbol{\Delta})^{-1}$), while $\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}(\mathbf{x}-\bar{\mathbf{x}}) = \boldsymbol{\Delta}\hat{\mathbf{F}}^{\text{wls}}$. ■

Regression scores shrink towards 0, reflecting the prior $\mathbf{F} \sim N(\mathbf{0}, \mathbf{I})$ (a Bayes estimate); WLS scores treat $\mathbf{F}$ as a fixed parameter and are conditionally unbiased.

5 PCA versus Factor Analysis

	PCA	Factor analysis
Goal	linear combinations explaining total variance $\text{tr}\boldsymbol{\Sigma}$	latent factors explaining covariances (off-diagonal)
Model	none (descriptive transformation)	$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$
Direction	variables $\to$ components ($\mathbf{Z} = \mathbf{U}^\top\mathbf{X}$)	factors $\to$ variables ($\mathbf{X} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}$)
Uniqueness	unique if eigenvalues distinct	only up to rotation
Increasing k	earlier components unchanged	(MLE) earlier loadings change

Chapter 10. Discrimination and Classification

Bayes rule, LDA/QDA, Fisher's discriminant, logistic regression, assessing classifiers

1 The Bayes Classification Rule

Motivation. Classification is a decision problem: assign a new observation to a group so as to minimize expected misclassification cost. The Bayes rule compares prior-weighted densities and is optimal; every practical classifier estimates it.

1.1 Setting

Let $Y \in \{1, \dots, K\}$ be the group and $\mathbf{X} \in \mathbb{R}^p$ the features, with priors $P(Y = k) = \pi_k$ and $\mathbf{X} \mid Y = k \sim f_k$. For $K = 2$: $\mathbf{X} \mid Y = 1 \sim f_1(\mathbf{x})$, $\mathbf{X} \mid Y = 2 \sim f_2(\mathbf{x})$, $P(Y = 1) = \pi_1$, and

$$\eta(\mathbf{x}) := P(Y = 1 \mid \mathbf{X} = \mathbf{x}) = \frac{f_1(\mathbf{x})\pi_1}{f_1(\mathbf{x})\pi_1 + f_2(\mathbf{x})\pi_2}.$$

The misclassification probability of a classifier δ (or ϕ) : $\mathbb{R}^p \to \{1, \dots, K\}$ is $P(\phi(\mathbf{X}) \neq Y) = \sum_k P(\phi(\mathbf{X}) \neq k \mid Y = k)P(Y = k)$.

Theorem 10.1: optimality of the Bayes rule

$\phi_{\text{Bayes}}(\mathbf{x}) = \arg\max_k P(Y = k \mid \mathbf{X} = \mathbf{x}) = \arg\max_k \pi_k f_k(\mathbf{x})$ minimizes the misclassification probability: $P(\phi_{\text{Bayes}}(\mathbf{X}) \neq Y) \leq P(\phi(\mathbf{X}) \neq Y)$ for every ϕ.

$K = 2$ **[Supplement].**

$$\begin{aligned}
\text{Risk}(\delta) &= \mathrm{E}[\mathbf{1}_{\{Y \neq \delta(\mathbf{X})\}}] = \mathrm{E}_Y\left[\mathrm{E}_{\mathbf{X}|Y}[\mathbf{1}_{\{Y \neq \delta(\mathbf{X})\}} \mid Y]\right] \\
&= \pi_1 \mathrm{E}_{\mathbf{X}|Y=1}[\mathbf{1}_{\{\delta(\mathbf{X}) \neq 1\}} \mid Y = 1] + \pi_2 \mathrm{E}_{\mathbf{X}|Y=2}[\mathbf{1}_{\{\delta(\mathbf{X}) \neq 2\}} \mid Y = 2] \\
&= \int \underbrace{\left[\pi_1 \mathbf{1}_{\{\delta(\mathbf{x}) \neq 1\}} f_1(\mathbf{x}) + \pi_2 \mathbf{1}_{\{\delta(\mathbf{x}) \neq 2\}} f_2(\mathbf{x})\right]}_{=:h(\mathbf{x})} d\mathbf{x}.
\end{aligned}$$

At each $\mathbf{x}$, $\delta(\mathbf{x}) = 1$ gives $h(\mathbf{x}) = \pi_2 f_2(\mathbf{x})$ and $\delta(\mathbf{x}) = 2$ gives $h(\mathbf{x}) = \pi_1 f_1(\mathbf{x})$. To minimize the integral choose the smaller pointwise: if $\pi_2 f_2(\mathbf{x}) < \pi_1 f_1(\mathbf{x})$ choose $\delta(\mathbf{x}) = 1$; if $>$, choose $\delta(\mathbf{x}) = 2$. Thus $\phi_{\text{Bayes}}(\mathbf{x}) = \arg\max_k P(Y = k \mid \mathbf{X} = \mathbf{x})$. For general K the same argument minimizes $1 - P(Y = \phi(\mathbf{x}) \mid \mathbf{x})$ pointwise. ■

Remark 10.2: misclassification costs

If $c(2 \mid 1)$ is the cost of classifying a member of group 1 into group 2, the rule minimizing expected cost classifies into group 1 when $\dfrac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \dfrac{c(1 \mid 2)}{c(2 \mid 1)} \dfrac{\pi_2}{\pi_1}$.

2 Normal Populations: LDA and QDA

2.1 Common covariance: linear discriminant analysis

If $f_k = N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$ ($\boldsymbol{\Sigma}_k = \boldsymbol{\Sigma}$), then $\log\{\pi_k f_k(\mathbf{x})\} = -\frac{1}{2}\mathbf{x}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}\boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \log \pi_k +$ const, and the first term does not depend on k, so

$$\phi(\mathbf{x}) = \arg\min_k \delta_k(\mathbf{x}), \qquad \delta_k(\mathbf{x}) = b_{0k} - \mathbf{b}_k^\top \mathbf{x}, \quad b_{0k} = \tfrac{1}{2}\boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k - \log(\pi_k), \quad \mathbf{b}_k = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k.$$

The discriminant function is linear in $\mathbf{x}$: **linear discriminant analysis (LDA)**. For $K = 2$, $\delta_1 < \delta_2$ rearranges to

$$\phi(\mathbf{x}) = 1 \;\text{ if }\; \mathbf{b}^\top\Big(\mathbf{x} - \frac{\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2}{2}\Big) < \log\Big(\frac{\pi_1}{\pi_2}\Big), \qquad \mathbf{b} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1).$$

$K = 2$ **[Supplement].** $\delta_1 < \delta_2 \iff (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}(\boldsymbol{\mu}_1^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2) > \log(\pi_2/\pi_1)$. Since $\boldsymbol{\mu}_1^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)$, this is $-\mathbf{b}^\top(\mathbf{x} - \frac{\boldsymbol{\mu}_1+\boldsymbol{\mu}_2}{2}) > \log(\pi_2/\pi_1)$. ■

The sample version plugs in $\boldsymbol{\mu}_k \to \bar{\mathbf{x}}_k$, $\boldsymbol{\Sigma} \to \mathbf{S}_p$ (pooled covariance), $\pi_k \to n_k/n$: $\phi(\mathbf{x}) = \arg\max_k(b_{0k} - \mathbf{b}_k^\top\mathbf{x})$-type rule with $\hat{\mathbf{b}}_k = \mathbf{S}_p^{-1}\bar{\mathbf{x}}_k$, and for $K = 2$, $\hat{\mathbf{b}} = \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$.

2.2 Different covariances: quadratic discriminant analysis

If $f_k = N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$,

$$\phi(\mathbf{x}) = \arg\max_k\Big\{-\tfrac{1}{2}\log|\boldsymbol{\Sigma}_k| - \tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^\top\boldsymbol{\Sigma}_k^{-1}(\mathbf{x} - \boldsymbol{\mu}_k) + \log\pi_k\Big\} = \arg\max_k\,(b_{0k} - \mathbf{b}_k^\top\mathbf{x} + \mathbf{x}^\top\mathbf{C}_k\mathbf{x})$$

with $\mathbf{C}_k = -\frac{1}{2}\boldsymbol{\Sigma}_k^{-1}$. The boundaries are quadratic surfaces: **QDA**. It needs $K \cdot p(p+1)/2$ more parameters, so with small samples LDA is often more stable (bias–variance trade-off).

3 Fisher's Linear Discriminant

Motivation. Fisher's approach needs no distributional assumption: find the projection that best separates the groups relative to the within-group spread. Under normality with equal covariances it coincides with the Bayes rule.

Without distributional assumptions, find the projection that best separates the groups.

Definition 10.3: Fisher's LDA direction ($K = 2$)

$$\mathbf{u}_0 = \arg\max_{\mathbf{u}\in\mathbb{R}^p} \frac{(\mathbf{u}^\top\bar{\mathbf{x}}_1 - \mathbf{u}^\top\bar{\mathbf{x}}_2)^2}{\mathbf{u}^\top\mathbf{S}_p\mathbf{u}}:$$

squared difference of projected means over projected within-group variance.

Theorem 10.4: solution

$\mathbf{u}_0 \propto \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)$ (equivalently $\mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$), and the maximum is $\mathbf{d}^\top\mathbf{S}_p^{-1}\mathbf{d}$ with $\mathbf{d} = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$, the squared sample Mahalanobis distance.

Proof [Supplement]. In general, to maximize $\frac{\mathbf{u}^\top\mathbf{A}\mathbf{u}}{\mathbf{u}^\top\mathbf{B}\mathbf{u}}$ ($\mathbf{B} \succ 0$) use $L(\mathbf{u}, \lambda) = \mathbf{u}^\top\mathbf{A}\mathbf{u} - \lambda(\mathbf{u}^\top\mathbf{B}\mathbf{u} - 1)$: $\partial L/\partial\mathbf{u} = \mathbf{0} \Rightarrow \mathbf{A}\mathbf{u} = \lambda\mathbf{B}\mathbf{u} \Rightarrow \mathbf{B}^{-1}\mathbf{A}\mathbf{u} = \lambda\mathbf{u}$, so the top eigenvector of $\mathbf{B}^{-1}\mathbf{A}$ is optimal. With $\mathbf{A} = \mathbf{d}\mathbf{d}^\top$ (rank 1) and $\mathbf{B} = \mathbf{S}_p$, $\mathbf{S}_p^{-1}\mathbf{d}\mathbf{d}^\top$ has a single nonzero eigenvalue $\mathbf{d}^\top\mathbf{S}_p^{-1}\mathbf{d}$ with unique eigenvector $\mathbf{S}_p^{-1}\mathbf{d}$ (rank 1). Directly: with $\mathbf{v} = \mathbf{S}_p^{1/2}\mathbf{u}$, Cauchy–Schwarz gives $(\mathbf{u}^\top\mathbf{d})^2/\mathbf{u}^\top\mathbf{S}_p\mathbf{u} = (\mathbf{v}^\top\mathbf{S}_p^{-1/2}\mathbf{d})^2/\mathbf{v}^\top\mathbf{v} \le \mathbf{d}^\top\mathbf{S}_p^{-1}\mathbf{d}$. ■

Remark 10.5: Fisher's rule = LDA with equal priors

With the discriminant score $y_i = \mathbf{b}_Y^\top\mathbf{x}_i$, $\mathbf{b}_Y = \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$, and the cut-off $b_{0Y} = \mathbf{b}_Y^\top(\bar{\mathbf{x}}_1 +$

$\bar{\mathbf{x}}_2)/2$,

$$\text{“}\mathbf{b}_Y^\top \mathbf{y} - b_{0Y} < 0 \Rightarrow 1\text{”} \iff \text{“}\big[\mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)\big]^\top \Big(\mathbf{x} - \frac{\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2}{2}\Big) < 0 \Rightarrow 1\text{”},$$

which is sample LDA with $\pi_1 = \pi_2$. A rule derived without normality coincides with the Bayes rule under normality.

Several groups ($K \geq 3$): $\max_{\mathbf{u}} \dfrac{\mathbf{u}^\top \mathbf{B}\mathbf{u}}{\mathbf{u}^\top \mathbf{W}\mathbf{u}}$ ($\mathbf{B}, \mathbf{W}$ the MANOVA between/within SSP matrices) is solved by the eigenvectors of $\mathbf{W}^{-1}\mathbf{B}$; there are at most $\min(p, K-1)$ nonzero eigenvalues, the same η_i that appear in the Wilks/Pillai statistics of Ch. 6.

4 Logistic Regression

For $K = 2$, $Y \in \{0, 1\}$, model the posterior directly:

$$\operatorname{logit}(\eta(\mathbf{x})) := \log\Big(\frac{\eta(\mathbf{x})}{1 - \eta(\mathbf{x})}\Big) = b + \boldsymbol{\beta}^\top \mathbf{x}, \qquad \eta(\mathbf{x}) = P(Y = 1 \mid \mathbf{X} = \mathbf{x}).$$

Classification: “$\eta(\mathbf{x}) > \frac{1}{2} \Rightarrow 1$” $\iff$ “$f(\mathbf{x}) = b + \boldsymbol{\beta}^\top \mathbf{x} > 0 \Rightarrow 1$” (a linear boundary). The log-likelihood is

$$\begin{aligned} \ell(b, \boldsymbol{\beta}) &= \sum_{i=1}^{n} \big[y_i \log(\eta(\mathbf{x}_i; b, \boldsymbol{\beta})) + (1 - y_i) \log\{1 - \eta(\mathbf{x}_i; b, \boldsymbol{\beta})\}\big] \\ &= \sum_{i=1}^{n} \big[y_i (b + \boldsymbol{\beta}^\top \mathbf{x}_i) - \log\{1 + \exp(b + \boldsymbol{\beta}^\top \mathbf{x}_i)\}\big], \end{aligned}$$

which is concave and maximized by Newton–Raphson (IRLS). In the equal-covariance normal model $\log \frac{\pi_1 f_1}{\pi_0 f_0}$ is exactly linear in $\mathbf{x}$, so the logistic model holds; LDA models the joint distribution, logistic regression only the conditional one.

Derivation: the second equality and the derivatives

$\log \eta = (b + \boldsymbol{\beta}^\top \mathbf{x}) - \log(1 + e^{b + \boldsymbol{\beta}^\top \mathbf{x}})$ and $\log(1 - \eta) = -\log(1 + e^{b + \boldsymbol{\beta}^\top \mathbf{x}})$, so $y \log \eta + (1 - y) \log(1 - \eta) = y(b + \boldsymbol{\beta}^\top \mathbf{x}) - \log(1 + e^{b + \boldsymbol{\beta}^\top \mathbf{x}})$. The gradient is $\sum_i (y_i - \eta_i)(1, \mathbf{x}_i^\top)^\top$ and the Hessian $-\sum_i \eta_i(1 - \eta_i)(1, \mathbf{x}_i^\top)^\top (1, \mathbf{x}_i^\top) \preceq 0$.

5 Simple Classifiers in High Dimensions

If $p > n$, $\mathbf{S}_p$ is singular and LDA cannot be used (e.g. the Golub data).

- **Nearest centroid**: LDA with $\boldsymbol{\Sigma} = \mathbf{I}$. $\phi(\mathbf{x}) = 1$ if $\mathbf{b}^\top(\mathbf{x} - \frac{\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2}{2}) < 0$, $\mathbf{b} = \bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1$ (assign to the closer mean).
- **Naive Bayes classifier (diagonal LDA)**: assume independent variables and replace $\mathbf{S}_p$ by $\mathbf{D}_p = \operatorname{diag}(\mathbf{S}_p)$: $\phi(\mathbf{x}) = 1$ if $\mathbf{b}^\top(\mathbf{x} - \frac{\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2}{2}) < 0$, $\mathbf{b} = \mathbf{D}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$.
- **k-nearest neighbors**: majority vote among the k training points closest to $\mathbf{x}$; non-parametric, high variance for small k and high bias for large k.

6 Assessment of Classifiers

- True error rate $P(\phi(\mathbf{X}) \neq Y) = \sum_{k=1}^{K} P(\phi(\mathbf{X}) \neq k \mid Y = k)P(Y = k)$; ϕ is optimal if $P(\phi(\mathbf{X}) \neq Y) \leq P(\varphi(\mathbf{X}) \neq Y)$ for all φ.
- Empirical misclassification rate $\hat{P}(\hat{\phi}_D(\mathbf{X}) \neq Y \mid D) = \frac{1}{M} \sum_{i=1}^{M} \mathbf{1}_{\{\hat{\phi}(\mathbf{x}_i) \neq y_i\}}$.
- **Training and test sets**: build $\hat{\phi}$ from $\mathbf{X}_{\text{tr}}$ and estimate the misclassification rate on $\mathbf{X}_{\text{test}}$. The training error (APER) is optimistically biased.
- **Confusion matrix**: actual group by predicted group (training and testing confusion matrices).
- **K-fold cross-validation**: split into K parts, use each in turn as the test set and average. $K \uparrow \Rightarrow$ bias $\downarrow$, variance $\uparrow$; $K \downarrow \Rightarrow$ the reverse. $K = n$ is LOOCV (Lachenbruch's holdout).

Chapter 11. Cluster Analysis

K-means, K-medoids, hierarchical clustering, Gaussian mixtures and the EM algorithm, choosing K

Unlike classification (Ch. 10), clustering groups similar observations without labels (unsupervised learning). A clustering is an encoder $C : \{1, \dots, n\} \to \{1, \dots, K\}$.

1 Dissimilarity and Within-Cluster Scatter

Motivation. Clustering looks for groups without labels, so we must decide what makes a grouping good. Small within-cluster scatter is the usual criterion; because total scatter is fixed, this is the same as large between-cluster scatter.

Definition 11.1: within-cluster scatter

For a dissimilarity $d(i, j) \geq 0$,

$$W(C) = \frac{1}{2} \sum_{k=1}^{K} \frac{1}{n_k} \sum_{i:C(i)=k} \sum_{j:C(j)=k} d(i, j), \qquad n_k = \#\{i : C(i) = k\}.$$

Smaller W is better.

Theorem 11.2: with squared Euclidean distance $W(C)$ is the within-cluster sum of squares

If $d(i, j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2$, then

$$W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|_2^2, \qquad \bar{\mathbf{x}}_{(k)} = \frac{1}{n_k} \sum_{C(i)=k} \mathbf{x}_i,$$

and $T = \sum_i \|\mathbf{x}_i - \bar{\mathbf{x}}\|^2 = W(C) + B(C)$ with $B(C) = \sum_k n_k \|\bar{\mathbf{x}}_{(k)} - \bar{\mathbf{x}}\|^2$.

Proof [Supplement]. Within cluster k, $\mathbf{x}_i - \mathbf{x}_j = (\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}) - (\mathbf{x}_j - \bar{\mathbf{x}}_{(k)})$ and $\sum_{C(i)=k} (\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}) = \mathbf{0}$, so

$$\sum_{C(i)=k} \sum_{C(j)=k} \|\mathbf{x}_i - \mathbf{x}_j\|^2 = 2n_k \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2;$$

multiply by $\frac{1}{2} \cdot \frac{1}{n_k}$. $T = W + B$ is the trace of the MANOVA SSP decomposition (Ch. 6). ∎

Since T is fixed, minimizing W is maximizing B.

1.1 Limits of combinatorial algorithms

The number of ways to partition n items into K non-empty clusters (Stirling number of the second kind) is

$$A(n, K) = \frac{1}{K!} \sum_{k=1}^{K} (-1)^{K-k} \binom{K}{k} k^n \sim \frac{K^n}{K!}.$$

E.g. $A(19, 4) \approx 10^{10}$: exhaustive search is impossible, so greedy iterative algorithms are used.

2 K-means and K-medoids

Motivation. Minimizing within-cluster scatter exactly is combinatorially hard. K-means alternates two easy steps, assigning points and updating centres, each of which decreases the criterion, so it converges to a local optimum.

2.1 K-means

With $d(i,j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2$, any point $\boldsymbol{m}$ satisfies $\sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{m}\|^2 \geq \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2$, so

$$W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|_2^2 = \min_{\boldsymbol{m}_1,\ldots,\boldsymbol{m}_K} L(C \mid \boldsymbol{m}), \qquad L(C \mid \boldsymbol{m}) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{m}_k\|_2^2.$$

Definition 11.3: K-means algorithm

(i) Minimize over C with $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$ fixed: update $C(i) = k$ for the closest center.

(ii) Minimize over $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$ with C fixed: update $\boldsymbol{m}_k$ by the new average of cluster k.

(iii) Iterate (i), (ii) until $L(C \mid \boldsymbol{m})$ does not change.

Theorem 11.4: convergence

K-means stops after finitely many iterations at a local minimum.

Proof [Supplement]. Each step minimizes L in one block of variables with the other fixed, so L is non-increasing. There are finitely many encoders C and step (ii) is unique for a given C, so while L strictly decreases no C can recur; the algorithm stops in finitely many steps. ■

The result depends on the starting values, so run several starts and keep the smallest W. K-means is scale-dependent; consider standardizing.

2.2 K-medoids

Use an arbitrary dissimilarity and restrict centers to observations.

(i) Minimize over C: update $C(i) = k$ for the closest medoid $\boldsymbol{m}_k$.

(ii) Minimize over $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$: update $\boldsymbol{m}_k$ by the item in the kth cluster that minimizes the total dissimilarity to the cluster.

(iii) Iterate (i), (ii) until $L(C \mid \boldsymbol{m})$ does not change.

Using actual observations as representatives makes it robust to outliers and usable for categorical data.

3 Hierarchical Clustering

3.1 Agglomerative

Start with singletons and repeatedly merge the two closest clusters. Between-cluster distances (linkage):

Single linkage	$d(G_1, G_2) = \min_{i \in G_1, j \in G_2} d_{ij}$
Complete linkage	$d(G_1, G_2) = \max_{i \in G_1, j \in G_2} d_{ij}$
Average linkage	$d(G_1, G_2) = \text{average}_{i \in G_1, j \in G_2} d_{ij}$

Single linkage tends to produce long chained clusters, complete linkage compact round ones. The result is drawn as a dendrogram and cut at a chosen height.

Distances between observations:

$$p\text{-norm: } d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_p = \Big(\sum_{l=1}^{d} |x_l - y_l|^p \Big)^{1/p},$$

$$\text{standardized distance: } d^2(\mathbf{x}, \mathbf{y}) = \sum_{l=1}^{d} \frac{(x_l - y_l)^2}{s_l^2} = (\mathbf{x} - \mathbf{y})^\top \mathbf{D}^{-1} (\mathbf{x} - \mathbf{y}).$$

3.2 Divisive

Start from a single cluster and split repeatedly (e.g. $K = 2$ means at each step).

4 Gaussian Mixtures and the EM Algorithm

4.1 Model

$\mathbf{X}_1, \ldots, \mathbf{X}_n \overset{\text{iid}}{\sim} g(\mathbf{x})$ with latent labels $\delta_1, \ldots, \delta_n \overset{\text{iid}}{\sim}$, $P(\delta_i = k) = \pi_k$, and $\mathbf{X} \mid \delta = k \sim N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$:

$$\mathbf{X} \sim g(\mathbf{x}) = \sum_{k=1}^{K} \text{pdf}(\mathbf{x} \mid \delta = k) P(\delta = k) = \sum_{k=1}^{K} \pi_k g_k(\mathbf{x}),$$

$$L(\boldsymbol{\theta} \mid \mathbf{X}) = \prod_{i=1}^{n} g(\mathbf{x}_i) = \prod_{i=1}^{n} \Big\{ \sum_{k=1}^{K} g_k(\mathbf{x}_i) \pi_k \Big\},$$

giving $\hat{\pi}_k, \hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k$. Then $\hat{P}(\delta = k \mid \mathbf{x}) = \hat{g}_k(\mathbf{x}) \hat{P}(\delta = k) / \hat{g}(\mathbf{x})$ and $C : \mathcal{X} \to \{1, \ldots, K\}$, $C(\mathbf{x}) = k$ iff $\hat{P}(\delta = k \mid \mathbf{x}) \geq \hat{P}(\delta = j \mid \mathbf{x})$ for all j (soft clustering). With all $\boldsymbol{\Sigma}_k = \sigma^2 \mathbf{I}$ and $\sigma \to 0$ this becomes K-means.

4.2 The EM algorithm ($K = 2$)

$\mathbf{X}_1, \ldots, \mathbf{X}_n \sim \pi_1 \phi(\mathbf{x}; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \pi_0 \phi(\mathbf{x}; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, $\pi_0 = 1 - \pi_1$. The observed-data log-likelihood

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \log \{ \pi_0 \phi(\mathbf{x}_i; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) + \pi_1 \phi(\mathbf{x}_i; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) \}$$

has no closed-form maximizer (log of a sum). If $\delta_1, \ldots, \delta_n \in \{0, 1\}$ were observed, $L(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n) = \prod_i \text{pdf}(\mathbf{x}_i, \delta_i \mid \boldsymbol{\theta}) = \prod_i \text{pdf}(\mathbf{x}_i \mid \delta_i, \boldsymbol{\theta}) \text{pdf}(\delta_i)$ and

$$\ell(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n) = \sum_{i : \delta_i = 1} \log \pi_1 \phi(\mathbf{x}_i; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \sum_{i : \delta_i = 0} \log \pi_0 \phi(\mathbf{x}_i; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$$

$$= \sum_{i=1}^{n} \left[\delta_i \log \pi_1 \phi_1(\mathbf{x}_i) + (1 - \delta_i) \log \pi_0 \phi_0(\mathbf{x}_i) \right],$$

which separates and gives closed forms such as $\hat{\pi}_1^{\text{MLE}} = n_1 / n$.

Definition 11.5: EM algorithm

(i) $\ell(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n)$ has an easy MLE, but $\delta_1, \dots, \delta_n$ are unobserved;
(ii) iteratively approximate the maximum. At $\boldsymbol{\theta}^{(m)}$:

- **E-step**: predict δ_i by $\hat{\delta}_i = \mathrm{E}(\delta_i \mid \mathbf{x}_i, \boldsymbol{\theta}^{(m)})$, i.e. compute $Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) := \mathrm{E}[\ell(\boldsymbol{\theta} \mid D) \mid D_{\text{obs}}, \boldsymbol{\theta}^{(m)}]$. Since the complete-data log-likelihood is linear in δ_i,

$$\hat{\delta}_i = \frac{\pi_1^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_1^{(m)}, \boldsymbol{\Sigma}_1^{(m)})}{\pi_0^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_0^{(m)}, \boldsymbol{\Sigma}_0^{(m)}) + \pi_1^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_1^{(m)}, \boldsymbol{\Sigma}_1^{(m)})} \quad \text{(responsibility)}.$$

- **M-step**: $\boldsymbol{\theta}^{(m+1)} = \arg\max_{\boldsymbol{\theta}} L(\boldsymbol{\theta} \mid \mathbf{X}^n, \hat{\boldsymbol{\delta}}^n)$, i.e. $\arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$:

$$\pi_1 = \frac{\sum \hat{\delta}_i}{n}, \quad \boldsymbol{\mu}_1 = \frac{\sum \hat{\delta}_i \mathbf{x}_i}{\sum \hat{\delta}_i}, \quad \boldsymbol{\Sigma}_1 = \frac{\sum \hat{\delta}_i (\mathbf{x}_i - \boldsymbol{\mu}_1)(\mathbf{x}_i - \boldsymbol{\mu}_1)^\top}{\sum \hat{\delta}_i},$$

  and similarly for component 0 with weights $1 - \hat{\delta}_i$.

Repeat until $d(\boldsymbol{\theta}^{(m)}, \boldsymbol{\theta}^{(m+1)})$ is small.

Theorem 11.6: monotonicity of EM

$\ell(\boldsymbol{\theta}^{(m+1)}) \geq \ell(\boldsymbol{\theta}^{(m)})$.

Proof [Supplement]. $\ell(\boldsymbol{\theta}) = Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$ with $H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = \mathrm{E}[\log p(\boldsymbol{\delta} \mid \mathbf{X}, \boldsymbol{\theta}) \mid \mathbf{X}, \boldsymbol{\theta}^{(m)}]$. By Gibbs' (Jensen's) inequality $H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) \leq H(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})$ (KL divergence ≥ 0). The M-step does not decrease Q, so $\ell(\boldsymbol{\theta}^{(m+1)}) - \ell(\boldsymbol{\theta}^{(m)}) = \{Q(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) - Q(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})\} - \{H(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})\} \geq 0$. ■

EM may converge to a local maximum (or saddle point), and the likelihood is unbounded as some $\boldsymbol{\Sigma}_k$ becomes singular, so use several starts and regularization.

5 Choosing K and Other Topics

- **Scatter decomposition**: $W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2$, $B(C) = \sum_{k=1}^{K} n_k \|\bar{\mathbf{x}}_{(k)} - \bar{\mathbf{x}}\|^2$. W always decreases as K grows, so it cannot choose K by itself (elbow method: where the decrease flattens).
- Indices: $\mathrm{CI}(K, c) = W(K, c)/T$ (smaller is better) and Calinski–Harabasz $\mathrm{CH}(K, c) = \dfrac{B(K, c)/(K-1)}{W(K, c)/(n-K)}$ (larger is better; the analogue of the one-way MANOVA F).
- **Gap statistic**: compare with the expected $\log W_K^*$ under a reference distribution (uniform over the data range),

$$\mathrm{Gap}(K) = \mathrm{E}^*[\log W_K^*] - \log W_K, \qquad s(K) = \text{(adjusted) s.d. of } \log W_K^*,$$

  and choose the smallest K with $\mathrm{Gap}(K) \geq \mathrm{Gap}(K+1) - s(K+1)$.
- **Clustering variables**: define $d(v_i, v_j) = 1 - r_{ij}$ or $1 - r_{ij}^2$ and apply hierarchical clustering.